

Assignment 1: Planning, A*, 5D-A*

Deadline: 27.01.2026

- Read this assignment fully before starting.
- Submit the homework in groups of two. Follow the groups previously assigned to you from the beginning of the course.
- Your submission should be archived in a single .zip (or similar) format. Name your folder and corresponding archive in the following format: “planning_ex1_name1_name2.zip”.
- Ensure all answers are legible. Text should be typed. Figures should have a sufficiently high resolution.
- For the coding part of the assignment, you should submit all the necessary functions/files to make your program run. Do NOT include the results of your program, unless it is stated to do so in the assignment. Your code must be written in Python3 and run on any machine with the commands listed in the assignment.
- For questions asking for explanations/justifications, always also include the assumptions under which your stance is valid.
- If you have any concerns or questions, email Linn (linn.chong@uni-bonn.de) or Marco (marco.gadallah@smail.inf.h-brs.de). Or better, come for the in-person tutorial!
- Have fun and good luck!

Feedback

Submit this page as a separate pdf.

1. In total, how much time did you both spend on this homework? _____ hours
2. How difficult do you find the homework? If you choose (d), please give at least one example of which problem it is.
 - (a) Easy: I can finish all problems with ease.
 - (b) Challenging: This is the right level of difficulty. With some efforts, I can finish most problems successfully.
 - (c) Difficult: After significant efforts, I still have difficulty and cannot finish several problems successfully.
 - (d) Unsuitably difficult: The problems are too difficult. I do not have sufficient information from the course materials to work on them. For example, problem _____
3. How is the collaboration within your team working out?
 - (a) We are both happy with the teamwork.
 - (b) We are having serious problems with working together because _____.

Exercise 1: Uninformed Search (50 Points)

In this exercise, you will implement the uninformed search algorithm to find the shortest path from a start to a goal location.

The input to the search is:

- the discretized map of the environment M
- the start position n_{start}
- the goal position n_{goal}

The output is the sequence of positions $n = (x, y)$ from the start to the goal.

At each step, the robot is only allowed to move from one cell to one of the **four neighboring cells**: left, right, top, and bottom. This search is uninformed and does not use any heuristics. The cost of each step is defined by the function $g(n)$, which is the cost of going from n_{start} to n , which is equal to the number of movements taken, a.k.a., length of the path.

1. Building upon the code given in this assignment (scripts/assignment_1_astar.py), implement the function 'plan_path_uninformed' to plan a path using the aforementioned search algorithm. We give an example of how your function should be called in 'scripts/example_test_case.py'. If there is no valid path, return an empty path.
2. What are the assumptions did you make when implementing the algorithm? Give at least two assumptions.
3. Is this algorithm complete? Why or why not?
4. Is this algorithm optimal? Why or why not?
5. What are the time and space complexities of the algorithm? Give some calculations on how you came to this answer.

Exercise 2: A* Algorithm (30 Points)

Improve upon the planner from the previous exercise by add a heuristic to speed up the search.

1. What is the condition the heuristic must meet in order to be admissible? Suggest a suitable simple heuristic for this use case (you can assume the search space is Euclidean in all tests).
2. Building upon the code given in this assignment (scripts/assignment_1_astar.py), implement the function 'plan_path_astar' to plan a path using the search algorithm with your heuristic. We give an example of how your function should be called in 'scripts/example_test_case.py'.
3. Is this algorithm complete? Why or why not?
4. Is this algorithm optimal? Why or why not?
5. What are the time and space complexities of the algorithm? Give some calculations on how you came to this answer.
6. What changes if M is dynamic, i.e., if the map changes with time? Are there changes to the algorithm's completeness, optimality or complexities? Why or why not?

Exercise 3: Further Improvements (20 Points)

1. What can you change to further speed up the planning algorithm (i.e., reduce time taken to plan a path)? What (new) assumption have you made? You can decide the assumption on your own, within reasonable means (e.g., you cannot assume that the map is always empty, but you can change the robot design a little.). Under this new assumption, is your algorithm optimal?
2. Implement your suggested changes as in the same script (scripts/assignment_1_astar.py) as the function 'plan_path_fast'.
3. Write a short report (max 2 pages in total) about your improvements. Include experiments to highlight the improvements of this new method. You can include some figures for qualitative results. Include all the maps you used in your experiments in your final submission such that we can replicate your results.